# Betteries

**OVER THE PAST 18 YEARS,** we've seen technology evolve rapidly, and consistently, so much so that I think if you were Rip Van Winkle-d over the past 20 years, you'd wake up in awe, or in sheer terror because of the advancements we've made. I dare say, it might arguably be the 20-year period in which we've made the most technological advancements, and the tech divide between generations is more massive than any time before.

There is however, one hugely important area where a Rip Van Winkle from the year 2000 would wake up and say, "Ah that's familiar!". If the headline didn't tip you off already, that area is battery technology, or more specifically better batteries – betteries . We at Digit have been complaining about batteries for ages, and in fact, older readers will know we've used this same headline in an article way back in June 2006 as well. Back then we were talking about fuel cells as a possible future battery technology. I need not tell you how that prediction turned out... While batteries have certainly got smaller, the real change is that devices have become less power hungry. It's an uneasy compromise, because if it were possible, we'd all love to be able to have desktop-like ray-tracing graphics in games on mobiles, but that level of processing would kill our batteries in seconds!

With all the concern over climate change, green power is important, and solar power is the cheapest and greenest power we have available. Since the sun will shine long after the earth is dead, solar power has no expiration date. The problem is, at night, when we come home from work and watch Netflix, run our LED lights and our air conditioning, the sun has already set, and we need batteries to store the energy it produced in the day time.

Tesla, and all the others playing catch up with it, are also struggling to improve battery tech drastically. With electric cars, it's a little more complex than just holding and releasing charge on a daily basis – the batteries have to charge really quickly, discharge slowly, and last decades for the cars to be viable to the masses.

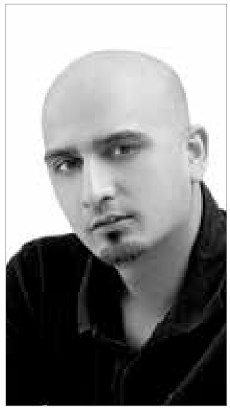
Batteries invade our lives far more now than ever before, and yet, they're essentially the same design that was thought up in 1991 – before our Rip Van Winkle went to sleep! The only thing I use from the 90s still is a stereo system with a cassette player, a CD player, and even a vinyl record player, and I use it to play my old collection of music, but not because it's superior quality... it's just considered vintage and nostalgic.

What's being worked on in labs across the globe? Pretty much mostly improvements to current Lithium-ion batteries. One major improvement that is being researched is changing the liquid substrate that the ions pass through to a solid. These solid state batteries will be a huge improvement in terms of safety, but not performance, not yet. Theoretically, however, if you have a solid substrate, the volume that the battery occupies goes down considerably, which means, either you get smaller and lighter batteries, or you get batteries with more range/running time in the same volume. Now, all they have to do is get them working in real-world conditions, and not just in ideal lab conditions.

Tesla researchers released a research paper recently that suggests they are able to achieve 8,000 charging cycles on single crystal, NMC532/AG cells, which is essentially over 20 years of a once daily charge cycle. Although it's in ideal lab conditions at low temperatures (20°C), even if it is half as efficient, it's still a huge improvement over current battery tech. The NMC532/AG stands for Li-Ni(0.5)-Mn(0.3)-Co(0.2)-O2/Artificial Graphite (Nickel, Manganese, Cobalt, in amounts of 0.5%, 0.3% and 0.2% respectively).

Then there's a possible shift away from lithium being researched. To be honest, this has been a research area for decades now, without any luck so far, but researchers are always hopeful, and are looking to replace lithium with something cheap and readily available. Part of the problem with lithium is sourcing it from across the globe, trade restrictions, and embargos, and the like. Now, if lithium could be replaced with something really cheap and easily available, such as sodium or silicon, then we could end up with really cheap batteries. We could even live with batteries not being as efficient as lithium batteries, because the significantly lower costs would allow us to stack more batteries in to make up for the reduced power, and still save loads of money. Another element being looked at more recently is calcium, which is also very abundant in nature, and cheap to produce.

I have my fingers crossed, hoping that we will see a much needed significant improvement in battery technology in under 5 years. History shows that all predictions about battery tech end up being wrong, and so five years from now you will probably find no changes whatsoever... Keep an eye out for future articles in Digit about 'betteries'.



**"Batteries invade our lives far more than before now, and yet, they're essentially the same design that was thought up in 1991"**

**Liked or hated this column? Let me know at:** editor@digit.in | facebook.com/raaabo | @raaabo | t.me/digitgeek | discord.gg/mYD8XVt